

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

Answer

- A. $23x(x^2 + 2x + 9)$
- B. $9x(23x^3 + 2x^2 + 1)$
- C. $x(23x^2 + 2x + 9)$
- D. $34(x^3 + x^2 + x)$

Correct Answer: C

Rationale

Choice C is correct. Since x is a common factor of each term in the given expression, the given expression can be rewritten as $x(23x^2 + 2x + 9)$.

Choice A is incorrect. This expression is equivalent to $23x^3 + 46x^2 + 207x$.

Choice B is incorrect. This expression is equivalent to $207x^4 + 18x^3 + 9x$.

Choice D is incorrect. This expression is equivalent to $34x^3 + 34x^2 + 34x$.